# The half adjoint equivalence structure on a map is a mere proposition

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

*Editorial note.* The counted statement is the author thm environment at source lines 359-361 and its proof environment opens on line 362, the next line; the pairing is by adjacency and the retained proof range 362-367 is the whole environment.

*Editorial note.* The retained counter values (chapter 4, section 2, theorem offset 12) make the counted theorem print as Theorem 4.2.13, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:hae-hprop = 4.2.13).

*Editorial note.* Four labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: ex:prop-inhabcontr (book Example 3.5, defined at line 1157 of the logic chapter pinned file), ex:sigma-assoc (book Example 2.10, whose chapter file is not part of the pinned corpus so the pointer span is the label line of this chapter dump), lem:inv-hprop (book Lemma 4.2.9, defined at line 319 of this chapter pinned file and counted by its own card of this lane) and lem:coh-hprop (book Lemma 4.2.12, defined at line 351 of the same file).

*Editorial note.* The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

*Editorial scope.* One original result of the pinned HoTT Book source tree, counted once: the half adjoint equivalence structure is a mere proposition (author label thm:hae-hprop, book Theorem 4.2.13).

*Allowed standard prerequisites (editorial).*

- Half adjoint equivalence structure. The type of half adjoint equivalences on f collects an inverse, a homotopy and a coherence path between the two composites.. It is the type shown to be a mere proposition.
- Being a proposition via contractibility. It suffices to assume the structure inhabited and prove the type contractible, by the standard criterion for propositionality.. It is the reduction the proof opens with.
- Associativity of the Sigma type. A Sigma type whose second component is itself a Sigma type is equivalent to the flattened Sigma type over the two components.. It rewrites the structure into a Sigma type over the right inverse type.

*Source reference.* Original remark 3.5: sources/hott-logic/pinned-original-source.tex, source lines 1157–1157. The book Example 3.5, the criterion that a type is a proposition as soon as it is contractible under an assumed inhabitant, used at the first line of the retained proof. Navigation pointer only.

*Source reference.* Original remark 2.10: sources/hott-equivalences/book-labelnumbers-first-edition.tex, source lines 1402–1402. The book Example 2.10, associativity of the Sigma type, used to flatten the structure. Navigation pointer only: the chapter file that carries it is not part of the pinned corpus, so the pointer span is the label line of this chapter dump.

*Source reference.* Original lemma 4.2.9: sources/hott-equivalences/pinned-original-source.tex, source lines 319–319. Book Lemma 4.2.9, contractibility of the two inverse types, cited for the first layer of the flattened structure. Navigation pointer only: it is counted by its own card of this lane.

*Source reference.* Original lemma 4.2.12: sources/hott-equivalences/pinned-original-source.tex, source lines 351–351. Book Lemma 4.2.12, contractibility of the coherence type, cited for the second layer. Navigation pointer only.

**Theorem 4.2.13.** *For any $f : A \to B$, the type* $\mathsf{ishae}(f)$ *is a mere proposition.*

*Proof.* By original remark 3.5 it suffices to assume $f$ to be a half adjoint equivalence and show that $\mathsf{ishae}(f)$ is contractible. Now by associativity of $\Sigma$ (original remark 2.10), the type $\mathsf{ishae}(f)$ is equivalent to

$$\sum_{u:\mathsf{rinv}(f)} \mathsf{rcoh}_f(\mathsf{pr}_1(u), \mathsf{pr}_2(u)).$$

But by original lemmas 4.2.9 and 4.2.12 and the fact that $\Sigma$ preserves contractibility, the latter type is also contractible. □